Market Expectations in the Cross-Section of Leading Present Values*

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October 12, 2026

Abstract: We revisit the cross-sectional present-value framework of Kelly and Pruitt (2013) with heterogeneous, time-varying information precision. Incomplete news incorporation can generate common valuation errors that violate weak-dependence assumptions for factor extraction. Guided by the model, we combine return-based Leader identification with supervised factor extraction from their valuations, which we call *leading present values*. Over 1980–2025, Negative-Leader forecasts of monthly U.S. stock market returns achieve an out-of-sample R^2 of 1.54% relative to the historical mean, outperforming full-panel valuation forecasts and forecasts using the value-weighted return of the same Leaders as the predictor. These findings demonstrate how integrating economic structure with supervised representation learning can improve forecast accuracy and economic value, providing a broader framework for economically informed machine learning in asset pricing.

Keywords: Return Predictability; Supervised Representation Learning; Present-Value Models; Information Precision; Leader Selection

JEL classification: C58; G12; G17.

1 Introduction

Measuring market expectations is central to asset pricing and portfolio choice, yet predicting aggregate stock returns remains difficult. A large literature uses aggregate valuation

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ratios and financial variables to forecast the equity premium (see, e.g., Fama and French 1988; Campbell and Shiller 1988*a*; Lettau and Ludvigson 2001; Welch and Goyal 2008; Rapach et al. 2016, and the references therein). Aggregation, however, can obscure distinct signals in the cross-section of asset prices, motivating the use of disaggregated prices and valuations to recover aggregate expectations (Polk et al. 2006; Kelly and Jiang 2014; Huang et al. 2015; Faias 2023). Machine learning provides tools for extracting signals from many correlated predictors while controlling noise and estimation error (Gu et al. 2020; Kozak et al. 2020; Giglio et al. 2022; Christensen et al. 2023; Chen, Pelger and Zhu 2024). The econometric challenge is to combine this flexibility with economic models of how prices reflect expectations.

Kelly and Pruitt (2013) provide a cross-sectional present-value framework for addressing this problem. Their system links individual valuation ratios to aggregate return and cash-flow expectations through common latent factors. Using the three-pass regression filter (3PRF) of Kelly and Pruitt (2015), they extract a predictive factor from the cross-section of book-to-market ratios and document strong return predictability in their original sample. In hindsight, this framework can be viewed as an influential application of supervised representation learning in asset pricing: the forecasting target guides factor extraction, separating predictive components from common variation unrelated to future market returns. However, Goyal et al. (2024) report a negative out-of-sample R^2 in an extended evaluation and conclude that *"the extended evidence should now be viewed as somewhat less consistent."* This deterioration raises a methodological question: how can supervised representation learning within the cross-sectional present-value framework deliver more reliable out-of-sample forecasts through economically guided selection of valuation inputs?

We propose an economically guided framework for supervised representation learning that combines return-based Leader identification with the three-pass regression filter of Kelly and Pruitt (2015). A present-value model with heterogeneous information precision provides the economic rationale for identifying Leaders and using their valuations for factor extraction. We refer to these valuations as *leading present values*. At each month-end, we estimate a

LASSO peer-return network using daily returns. The bottom and top deciles of return deviations on the final trading day identify Negative and Positive Leaders, respectively. We then apply 3PRF to their available valuation histories: short-horizon returns identify informative measurements, while valuations are used to estimate the persistent state governing expected market returns.

The forecasting gains are substantial. We use daily returns and monthly book-to-market ratios for 100 U.S. stock portfolios sorted by size and book-to-market from January 1930 through December 2025 to forecast monthly returns on the CRSP value-weighted index. In a strictly recursive out-of-sample exercise, 3PRF forecasts based on Negative Leader valuations achieve an R^2_{OOS} of 2.61% relative to the historical mean over January 1980–December 2010, compared with 0.93% for the Full Panel. When the evaluation extends through December 2025, the forecast based on Negative Leaders retains an out-of-sample R^2 of 1.54%, significant at the 1% level, while the R^2 of the full-panel forecast based on Kelly and Pruitt (2013) falls to -0.19%. For a mean–variance investor, the forecast also generates annualized gains of 2.19 percentage points in certainty-equivalent returns and 0.168 in the Sharpe ratio. The forecasting gains persist across the economic states examined and after major crises are excluded. Economic gains also survive trading costs and alternative risk aversion and portfolio weight specifications.

To explain why selection helps, we extend the present-value system of Kelly and Pruitt (2013) to allow information precision to differ across assets and vary over time. Incomplete incorporation of aggregate news can generate a pervasive common component in valuation errors, violating the weak-dependence assumption underlying standard consistency results for 3PRF. This dependence motivates selecting assets that incorporate news more precisely to reduce exposure to common measurement error. Ambiguity about signal quality (Epstein and Schneider 2008) induces stronger updating in response to unfavorable signals. Accounting for both responses to current news and delayed incorporation of earlier news, we establish sufficient conditions under which negative-tail selection identifies a higher expected proportion of assets with currently high precision than random or positive-tail selection. These

advantages persist as the cross-section grows. Simulations with rolling LASSO estimation illustrate the identification gains in finite panels.

External news measures constructed independently of selection and forecasting show that Negative Leaders respond more strongly to both favorable and unfavorable news than Positive Leaders and Nonleaders, supporting the information precision mechanism. Dynamic selection also outperforms fixed subsets based on characteristics and random monthly selection. Finally, a smaller effective pool of Negative Leaders is associated with larger full-panel forecast losses relative to both the historical mean and the forecast based on Negative Leaders. Together with the smaller average pool after 2010, this evidence helps explain the deterioration of full-panel forecasts in the extended sample.

Our study contributes to three strands of literature. We connect the cross-sectional present-value approach to research on gradual information diffusion and lead–lag relations in stock returns (Hou 2007; Cohen and Frazzini 2008; Menzly and Ozbas 2010). Our closest comparison is Chen, Hao, Yu and Zhou (2024), who identify leading stocks and use their portfolio returns to forecast market returns. We use a similar return-based approach to identify Leaders, but their valuations, rather than their returns, enter supervised factor extraction. The model motivates this distinction: short-horizon returns reflect revisions in state assessments, whereas valuation levels retain exposure to the persistent state governing expected returns. Using the same portfolio data and Leader sets at each date, we compare the resulting valuation forecasts with forecasts based directly on Leader returns. The return-based predictors have positive out-of-sample R^2 over 1980–2010, but Negative Leader valuations yield higher R^2 in both evaluation periods.

We are not the first to investigate how predictor selection can improve factor-based forecasting. Boivin and Ng (2006) show that predictor composition and cross-sectional error dependence matter for factor forecasts, while Bair et al. (2006) and Bai and Ng (2008) develop methods that screen predictors before extracting factors. More recently, Yousuf and Feng (2022) use partial distance correlation to screen predictors and apply their methods to factor-based return forecasting with portfolio valuation ratios. Huang et al. (2022) scale

predictors by their predictive slopes before applying PCA, while Giglio et al. (2026) combine iterative screening, factor extraction, and projection to recover predictive information from weak factors. We ground predictor screening in a structural present-value model with heterogeneous information precision. The model establishes conditions under which return deviations identify assets with currently high precision and explains how selecting their valuations mitigates common measurement error. This gives selection a clear economic interpretation and complements the use of the forecasting target to guide factor extraction in 3PRF.

Our framework contributes to the integration of economic structure and machine learning in empirical asset pricing. Machine learning can improve return forecasts (Gu et al. 2020), and related research demonstrates the value of combining statistical flexibility with economic structure (Kozak et al. 2020; Gu et al. 2021; Chen, Pelger and Zhu 2024). We use machine learning to estimate cross-sectional return relationships and the information model to guide selection for supervised factor extraction. When used to estimate peer returns for Leader identification, regularized linear models, tree-based methods, and neural networks all deliver positive out-of-sample R^2 and economic gains, indicating that the benefits extend across learning algorithms. The framework offers a broader role for machine learning in asset-return forecasting: it can estimate underlying economic relationships that allow economic structure to guide forecasts, rather than serving only as a direct prediction tool.

The remainder of the paper is organized as follows. Section 2 develops the model, and Section 3 presents Leader identification and the forecasting method. Section 4 describes the data and recursive empirical design. Section 5 reports the main results. Sections 6 and 7 examine economic mechanisms and robustness, respectively. Section 8 concludes.

2 A Cross-Sectional Present-Value Model with Heterogeneous Information

2.1 Present Values under Heterogeneous Information

Consider N assets indexed by $i = 1, \ldots, N$. For asset i, let $r_{i,t+1}$ denote the log return, $\Delta cf_{i,t+1}$ log cash-flow growth, and $v_{i,t} = \log(B_{i,t}/M_{i,t})$ the log book-to-market ratio, where book equity $B_{i,t}$ and market equity $M_{i,t}$ are positive. Following the cross-sectional present-value approach of Kelly and Pruitt (2013), we relate valuations to a persistent common state,

$$F_t = \lambda F_{t-1} + \xi_t, \qquad 0 < \lambda < 1, \tag{1}$$

where ξ_t is an independent standard normal innovation. The aggregate forecasting targets follow the reduced-form dynamics

$$\begin{aligned} r_{m,t+1} &= \gamma_0 + \gamma F_t + \eta^{m,r}_{t+1}, \\ \Delta cf_{m,t+1} &= \delta_0 + \delta F_t + \eta^{m,cf}_{t+1}. \end{aligned} \tag{2}$$

The aggregate innovations have zero conditional means and are orthogonal to information available at t. The same state therefore governs the predictable components of market returns and cash-flow growth.

To model how valuations incorporate information about this state, we assume that investors observe F_{t-1} but infer the current innovation ξ_t from noisy signals specific to each asset. Signals have one of two precision levels, allowing information quality to vary across assets and over time:

$$s_{i,t} = \xi_t + \sigma_{i,t}\varepsilon_{i,t}, \qquad \sigma_{i,t} \in \{\sigma_1, \sigma_2\}, \qquad 0 \le \sigma_1 < \sigma_2. \tag{3}$$

The asset-specific noises $\varepsilon_{i,t}$ are i.i.d. standard normal variables, independent of aggregate innovations. Here, information precision refers to the precision of the asset-specific signal about aggregate news: a lower noise variance $\sigma^2_{i,t}$ indicates higher precision. Each period, m assets, where $m \in \{1, \ldots, N-1\}$, have high information precision, with signal-noise variance σ_1^2, and the remaining assets have variance σ_2^2. Their identities follow a stationary process

independent of aggregate innovations and asset-specific noises: with probability $\eta \in (0,1)$, the high-precision set is retained; otherwise, a new set of m assets is drawn uniformly and independently.

This process captures information acquisition and processing advantages that persist but shift across assets. Such advantages may reflect differences in research effort, investor attention, or access to information, and may shift as these conditions change. Persistence allows an advantage revealed by current returns to remain relevant for subsequent news responses without making high precision a permanent asset characteristic.

Although investors do not know the true noise variance of $s_{i,t}$, they can assess a plausible range based on the expertise and information access of the source. For each information type $g \in \{1,2\}$, we assume that this range contains the true noise variance and has a common width:

$$0 \le \underline{\sigma}_g^2 \le \sigma_g^2 \le \overline{\sigma}_g^2, \qquad \overline{\sigma}_g^2 - \underline{\sigma}_g^2 = \overline{\chi} \ge 0, \qquad g \in \{1,2\},$$

with $\underline{\sigma}_1 < \underline{\sigma}_2$. Lower interval endpoints indicate more reliable sources, while the common variance-interval width $\overline{\chi}$ measures ambiguity about signal quality. Let $\underline{\sigma}_{i,t}$ and $\overline{\sigma}_{i,t}$ denote the interval endpoints for the information type of asset i at time t.

Investors consider candidate models χ of signal quality that assign each information type a Gaussian noise distribution with variance $\chi_{i,t} \in [\underline{\sigma}_{i,t}^2, \overline{\sigma}_{i,t}^2]$. For each possible noise variance, Bayesian updating gives the conditional expectation of F_t:

$$\mathbb{E}_\chi[F_t \mid \mathcal{I}_{i,t}] = \lambda F_{t-1} + k_{i,t}(\chi) s_{i,t}, \qquad k_{i,t}(\chi) = \frac{1}{1+\chi_{i,t}}. \tag{4}$$

Here, $k_{i,t}(\chi)$ is the signal weight, $\mathbb{E}_\chi$ denotes expectation under model χ, and $\mathcal{I}_{i,t}$ is the information used to price asset i at time t. A candidate model with a lower signal-noise variance assigns greater weight to the same observed signal and thus implies a larger revision to the assessment of the current state.

At the asset level, we assume that expected returns and cash-flow growth are affine in the state assessments of investors under each candidate model:

$$\begin{aligned} \mu_{i,t}(\chi) &= \mathbb{E}_\chi[r_{i,t+1}(\chi) \mid \mathcal{I}_{i,t}] = \gamma_{i,0} + \gamma_i \mathbb{E}_\chi[F_t \mid \mathcal{I}_{i,t}], \\ g_{i,t}(\chi) &= \mathbb{E}_\chi[\Delta cf_{i,t+1} \mid \mathcal{I}_{i,t}] = \delta_{i,0} + \delta_i \mathbb{E}_\chi[F_t \mid \mathcal{I}_{i,t}] + \nu_{i,t}. \end{aligned} \tag{5}$$

Here, $r_{i,t+1}(\chi)$ is the return implied by the valuation path under candidate model χ. Asset-specific loadings allow responses to differ across assets, and $\nu_{i,t}$ captures an asset-specific component of expected cash-flow growth. We connect these expectations to valuations by applying the log-linear present-value identity (Campbell and Shiller 1988*b*; Vuolteenaho 2002) within each candidate model:

$$v_{i,t}(\chi) = -\frac{\kappa_i}{1-\rho_i} + \sum_{j=1}^{\infty} \rho_i^{j-1} \mathbb{E}_\chi[r_{i,t+j}(\chi) - \Delta cf_{i,t+j} \mid \mathcal{I}_{i,t}], \tag{6}$$

where $0 < \rho_i < 1$ and κ_i are linearization constants. Cash-flow growth $\Delta cf_{i,t+1}$ is proxied by log ROE,

$$ROE_{i,t+j} = \log\left(1 + \frac{\text{earnings}_{i,t+j}}{\text{book equity}_{i,t+j-1}}\right).$$

Combining this weighted sum of expected one-period returns net of cash-flow growth across all future horizons with (5) and (1) yields the valuation relation under each candidate model:

$$v_{i,t}(\chi) = \phi_{i,0} - \phi_i \mathbb{E}_\chi[F_t \mid \mathcal{I}_{i,t}] + e^v_{i,t}, \tag{7}$$

where the expressions for $\phi_{i,0}$ and ϕ_i are provided in Internet Appendix A.1. The parameters $\phi_i > 0$ are stable over time but may differ across assets. The valuation disturbance $e^v_{i,t}$ captures the present-value contribution of asset-specific cash-flow expectations. It is stationary with finite variance and is independent of the common state and signal process.

Together with (2), this relation links valuation assessments to the common state governing expected market returns and cash-flow growth.[1] It provides a basis for extracting information about aggregate expectations from cross-sectional valuations while allowing their informativeness to vary with signal precision.

Following the treatment of ambiguity about signal quality in Epstein and Schneider (2008), we assume that ambiguity-averse investors use the candidate model that yields the lowest current market value. This is equivalent to choosing the highest book-to-market ratio, $v_{i,t} = \max_\chi v_{i,t}(\chi)$, over the admissible candidate models. Since $-\phi_i < 0$, the selected weight $k_{i,t}$ is smallest for favorable signals and largest for unfavorable signals:

[1]The model uses a scalar state to match the baseline empirical specification and can be extended to multiple predictive factors.

$$k_{i,t} = \begin{cases} k_{i,t}^{+} = (1 + \overline{\sigma}_{i,t}^{2})^{-1}, & s_{i,t} \geq 0, \\ k_{i,t}^{-} = (1 + \underline{\sigma}_{i,t}^{2})^{-1}, & s_{i,t} < 0. \end{cases} \tag{8}$$

For $\overline{\chi} > 0$, $k_{i,t}^{-} > k_{i,t}^{+}$: favorable news can be attributed to noise, whereas unfavorable news is treated as a more precise indication of adverse fundamentals. When $\overline{\chi} = 0$, signal quality is unambiguous. Investors then assign the same weight to favorable and unfavorable signals, and the valuation discount associated with ambiguity disappears.

Substituting the selected signal weight into (7) and using the state and signal equations yields the observed valuation and its measurement error:

$$\begin{aligned} v_{i,t} &= \phi_{i,0} - \phi_i F_t + \omega_{i,t}^{v}, \\ \omega_{i,t}^{v} &= \phi_i(1 - k_{i,t})\xi_t - \phi_i k_{i,t}\sigma_{i,t}\varepsilon_{i,t} + e_{i,t}^{v}. \end{aligned} \tag{9}$$

The measurement error consists of unincorporated aggregate news, $\phi_i(1 - k_{i,t})\xi_t$; noise in asset-specific signals, $-\phi_i k_{i,t}\sigma_{i,t}\varepsilon_{i,t}$; and the valuation-specific disturbance, $e_{i,t}^{v}$. Only the last component remains when signals are noiseless and unambiguous. Unincorporated news can be common to many assets, so its effect need not disappear when valuations are combined. This creates a measurement problem when extracting the common state governing aggregate expectations.

2.2 The Three-Pass Regression Filter and Cross-Sectional Dependence

Following Kelly and Pruitt (2013), we use the three-pass regression filter to extract the common component relevant for forecasting market returns. The standard consistency result under an approximate factor structure assumes weak cross-sectional dependence in valuation errors. After the unconditional error means are absorbed into the intercepts, Assumption 3.1(a) of Kelly and Pruitt (2015) requires a finite constant M such that

$$\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{N}\sup_{t,s}\left|\mathbb{E}(\omega_{i,t}^{v}\omega_{j,s}^{v})\right| \leq M. \tag{10}$$

This condition permits some cross-sectional correlation but rules out a pervasive common error.

Incomplete incorporation of aggregate news can generate a common component in valuation errors that violates weak cross-sectional dependence. To show why this component persists as the cross-section grows, we examine the covariance of the complete valuation error for each asset with the aggregate innovation, ξ_t. The Gaussian signal structure implies

$$\begin{aligned}\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{N}\left|\mathrm{Cov}(\omega_{i,t}^{v},\omega_{j,t}^{v})\right| &\geq N\,\mathrm{Var}\left(\frac{1}{N}\sum_{i=1}^{N}\omega_{i,t}^{v}\right)\\ &\geq N\left[\mathrm{Cov}\left(\frac{1}{N}\sum_{i=1}^{N}\omega_{i,t}^{v},\xi_t\right)\right]^{2}=\frac{a^{2}}{N}\left(\sum_{i=1}^{N}\phi_i\right)^{2},\end{aligned} \tag{11}$$

where $a=1-\frac{1}{2}\mathbb{E}[k_{i,t}^{-}+k_{i,t}^{+}]\geq\overline{\chi}/[2(1+\overline{\chi})]>0$ uniformly in N for any fixed ambiguity width $\overline{\chi}>0$. Internet Appendix A.2 provides the derivation.

Under full information, the common state captures shared exposure to aggregate news. With noisy signals and ambiguity about their quality, prices incorporate this news only partially, leaving a common component in valuation errors. If exposure to unincorporated news remains widespread and economically meaningful as the cross-section expands, $aN^{-1}\sum_i\phi_i$ stays bounded away from zero. This sufficient condition is economically natural for broad equity panels whose constituent portfolios retain systematic exposure to aggregate news. For example, unexpected changes in interest rates or corporate taxes can generate substantial valuation responses of the same sign across firms despite differences in their sensitivities. The lower bound then grows at least linearly with N, violating the weak-dependence condition (10).

Holding the valuation loading fixed, the lower noise-variance interval associated with higher information precision raises both updating weights and reduces the projection loading of the valuation error on aggregate news. This motivates selecting valuations that reflect the current predictive state more fully and reduce exposure to common measurement error that need not vanish as the panel expands. Information precision thus provides an economic criterion for selecting inputs to supervised factor extraction. Because precision is unobserved and can change over time, this criterion requires observable evidence of how assets incorporate current news. The next section develops a selection rule based on return deviations.

3 Leader Identification and Forecasting

3.1 From Information Precision to Leader Identification

In this subsection, t denotes a generic short interval; the empirical implementation uses daily observations. The log-linear return identity is

$$r_{i,t} \approx \kappa_i + v_{i,t-1} - \rho_i v_{i,t} + \Delta cf_{i,t}, \tag{12}$$

where ρ_i is typically close to one (e.g., 0.96 for annual data and 0.996 for monthly data; see Campbell and Shiller 1988*a*; Campbell 1991). At short horizons, cash-flow growth is small relative to valuation changes induced by prices (Cochrane 2011). Approximating ρ_i by one and cash-flow growth by its mean gives

$$r_{i,t} \approx c_i + v_{i,t-1} - v_{i,t}. \tag{13}$$

Substituting (9) into (13) yields

$$r_{i,t} \approx c_i + \phi_i(\lambda - 1)F_{t-1} + \phi_i k_{i,t}\xi_t + \phi_i(1 - k_{i,t-1})\xi_{t-1} + e^r_{i,t}, \tag{14}$$

$$e^r_{i,t} = \phi_i\left[k_{i,t}\sigma_{i,t}\varepsilon_{i,t} - k_{i,t-1}\sigma_{i,t-1}\varepsilon_{i,t-1}\right] + e^v_{i,t-1} - e^v_{i,t}.$$

Returns reflect both timely incorporation of current news, $\phi_i k_{i,t}\xi_t$, and delayed incorporation of preceding news, $\phi_i(1-k_{i,t-1})\xi_{t-1}$. This links the model to information-based explanations of return lead–lag relations. The disturbance $e^r_{i,t}$ also depends on the updating weights and need not be independent of aggregate news.

We use the short-horizon approximation in (13) as the return specification for a simplified benchmark that establishes how deviations from peer returns identify assets with higher current information precision. The benchmark assumes a common valuation loading, $\phi_i = \phi > 0$ in (9), and a common constant return component, $c_i = c$. The centered valuation disturbances $e^v_{i,t}$ are exchangeable: their joint distribution is unchanged when assets are relabeled. Define the asset–peer return deviation as

$$D_{i,t} = r_{i,t} - \sum_{j \neq i} w_{ij} r_{j,t}, \tag{15}$$

where $w_{ij} := \theta_{ij} / \sum_{\ell \neq i} |\theta_{i\ell}|$, and θ_{ij} are the population LASSO coefficients in the benchmark peer-return regression $r_{i,t} = \alpha_i + \sum_{j \neq i} \theta_{ij} r_{j,t} + u_{i,t}$. In the empirical implementation, this

normalization sets the gross exposure of each peer portfolio to one, facilitating comparisons across assets while preserving the signs and relative magnitudes of the estimated coefficients.

To assess whether extreme return deviations identify more informative assets as the cross-section grows, consider panels with $m_N = \lfloor pN \rfloor$ assets of currently high precision and select the $\ell_N = \lfloor \alpha N \rfloor$ lowest and highest deviations. The high-precision share $p \in (0, 1)$ and the tail share $\alpha \in (0, 1/2]$ remain fixed, as do $\overline{\chi}$ and η. Let $\pi_{-,N}$ and $\pi_{+,N}$ denote the expected proportions of currently high-precision assets in the lower and upper tails, respectively. Random selection has expected proportion $p_N = m_N/N \to p$.

Proposition 3.1. *Suppose $\overline{\chi} > 0$ and $p^2 < \alpha \leq p$. There exists a constant $\epsilon_0 \in (0, 1)$ such that the following statements hold whenever $0 < \sigma_1 < \epsilon_0$, $\epsilon_0^{-1} < \sigma_2 < \infty$, and $\mathrm{Var}(e^v_{i,t})/\phi^2 < \epsilon_0^2$ for all i, N:*

(i) If $p \leq 1/2$, negative-tail selection retains an advantage over random and positive-tail selection:

$$\liminf_{N\to\infty}(\pi_{-,N} - p_N) > 0, \qquad \liminf_{N\to\infty}(\pi_{-,N} - \pi_{+,N}) > 0.$$

(ii) If $p \leq 1/3$, positive-tail selection also retains an advantage over random selection:

$$\liminf_{N\to\infty}(\pi_{+,N} - p_N) > 0.$$

Remark 3.1. Proposition 3.1 gives sufficient conditions under which negative-tail selection identifies a higher expected proportion of currently high-precision assets than random or positive-tail selection, even with nonzero signal noise and valuation disturbances. These advantages do not vanish as the cross-section grows, motivating the use of negative return deviations to identify informative valuations. Both tails outperform random selection when high precision is sufficiently scarce. Because p is unknown in applications, the condition $\alpha \leq p$ links the choice of tail size to the prevalence of high-precision assets. Internet Appendix A.3 provides the proof. Corollary A.2 extends the result to heterogeneous valuation loadings and return intercepts, with the population peer regressions re-estimated.

These results motivate a two-stage forecasting procedure. The first stage identifies Negative and Positive Leaders from the lower and upper tails of asset–peer return deviations. The

second applies supervised factor extraction to their valuation histories to forecast market returns. We first describe estimation of the peer-return benchmark and construction of the Leader sets.

3.2 Stage 1: LASSO-Based Leader Identification

The identification argument relies on the short-horizon approximation in (13). We therefore identify Leaders using daily asset returns. At each forecast origin t, network estimation and Leader classification use only returns observed at t or earlier. The resulting set determines which monthly valuation histories enter factor extraction to predict $r_{m,t+1}$. Thus, the valuation cross-section is filtered using only information available at the forecast origin.

Let $\mathcal{W}_t$ denote the five-year window of daily observations ending on the final trading day of month t. For each target asset i, we estimate its contemporaneous relation with peer returns using LASSO:

$$(\widehat{\alpha}_{i,t}, \widehat{\boldsymbol{\theta}}_{i,t}) = \arg\min_{\alpha,\boldsymbol{\theta}} \left\{ \frac{1}{2|\mathcal{W}_t|} \sum_{d\in\mathcal{W}_t} \left(r_{i,d} - \alpha - \boldsymbol{\theta}'\mathbf{r}_{-i,d}\right)^2 + \lambda_{i,t}\|\boldsymbol{\theta}\|_1 \right\}. \tag{16}$$

Here, $|\mathcal{W}_t|$ is the number of daily observations in the estimation window, and the penalty parameter $\lambda_{i,t}$ is selected by five-fold blocked cross-validation.

For portfolio returns, the cross-section may be smaller than the time-series dimension, but strong collinearity among peers can still make a dense least-squares network unstable. For individual securities, the number of candidate peers can approach or exceed the number of time-series observations, so the ℓ_1 penalty also regularizes a high-dimensional regression. LASSO shrinks weak links, selects a sparse set of peers, and stabilizes the fitted benchmark (Tibshirani 1996; Chinco et al. 2019; Kozak et al. 2020; Babii et al. 2024).

We define the Return Deviation of asset i in month t as

$$RetDev_{i,t} = r_{i,t}^{\text{last}} - \sum_{j\neq i} \widehat{w}_{i,j,t} r_{j,t}^{\text{last}}. \tag{17}$$

Here, $r_{i,t}^{\text{last}}$ is the return on asset i on the final trading day of month t, and $\widehat{w}_{i,j,t}$ is the normalized LASSO coefficient:

$$\widehat{w}_{i,j,t} = \frac{\widehat{\theta}_{i,j,t}}{\sum_{k\neq i} |\widehat{\theta}_{i,k,t}|}.$$

This measure compares the latest asset return with the return on the estimated peer portfolio.

At the end of month t, we sort assets by $RetDev_{i,t}$. The Positive and Negative Leader sets are

$$\mathcal{L}_t^+ = \{\text{the top decile of } RetDev_{i,t}\}, \qquad \mathcal{L}_t^- = \{\text{the bottom decile of } RetDev_{i,t}\}. \tag{18}$$

The next stage uses the valuation ratios of the identified Leaders to forecast the next-period market return.

3.3 Stage 2: Factor Extraction and Prediction

The first stage selects assets whose return deviations indicate relatively precise incorporation of current information. We then apply the three-pass regression filter (3PRF) of Kelly and Pruitt (2015) to their valuation histories, using the forecasting target to guide extraction of the common predictive component. Selection determines which valuations enter the analysis, while supervised factor extraction determines how they are combined for forecasting.

At forecast origin T, let $\mathcal{L}_T$ denote the current Positive or Negative Leader set. Following Kelly and Pruitt (2013), we use realized one-period-ahead market returns in the estimation sample to identify the predictive component of the valuation panel.

Pass 1: Time-Series Regressions. For each $i \in \mathcal{L}_T$, estimate

$$v_{i,t} = \psi_{i,0} + \psi_i r_{m,t+1} + e_{i,t}, \qquad t = 1, \ldots, T-1. \tag{19}$$

The estimated slope $\widehat{\psi}_i$ measures the association between the valuation ratio of asset i and the next-period market return.

Pass 2: Cross-Sectional Regressions. For each month t, regress the valuation ratios of the current Leader set on the estimated first-pass slopes:

$$v_{i,t} = c_t + \widehat{F}_t \widehat{\psi}_i + \omega_{i,t}, \qquad i \in \mathcal{L}_T. \tag{20}$$

The cross-sectional slope $\widehat{F}_t$ summarizes the component of current valuations that predicts returns and serves as the extracted predictive factor.

Pass 3: Predictive Regression. Finally, estimate the predictive regression

$$r_{m,t+1} = \beta_0 + \beta \widehat{F}_t + \eta_{t+1}, \qquad t = 1, \ldots, T-1. \tag{21}$$

At forecast origin T, the corresponding real-time forecast is

$$\widehat{r}^{L}_{m,T+1|T} = \widehat{\beta}_0 + \widehat{\beta}\widehat{F}_T.$$

Remark 3.2. Using the full available history does not require currently selected assets to have been Leaders throughout the estimation period. Current return deviations identify assets with relatively precise information at the forecast origin, whereas historical regressions estimate the relationship between their valuations and subsequent market returns. In the model, valuation loadings are stable while information precision follows a stationary process. Past observations can therefore remain informative about this relationship even when an asset was not selected as a Leader. The first-pass regressions summarize valuation–return relationships across changes in information quality, and their estimated slopes are then used to combine the current valuations of selected assets into a predictive factor.

Remark 3.3. Chen, Hao, Yu and Zhou (2024) use the value-weighted return on leading assets to predict the next-period market return. Equation (14) offers an interpretation: greater information precision strengthens the current-news component of asset returns, allowing Leader returns to convey predictive information. Stronger responses to current news do not, however, ensure that these returns fully capture the persistent state. Returns also reflect delayed incorporation of earlier news and asset-specific disturbances. Under the short-horizon approximation, their direct loading on the lagged state F_{t-1} is $\phi_i(\lambda - 1)$, which is small when the state is highly persistent. Valuations, by contrast, retain a direct loading of $-\phi_i$ on F_t.

Let $r_{L,t}$ denote the Leader portfolio return and $R^2(x_t)$ the population R^2 from regressing $r_{m,t+1}$ on a constant and x_t. Under (2), with $\gamma \neq 0$, the population predictive R^2 satisfies

$$R^2(r_{L,t}) = \mathrm{Corr}^2(r_{L,t}, F_t)R^2(F_t) \leq R^2(F_t). \tag{22}$$

The inequality is strict unless Leader returns are perfectly correlated with F_t. It establishes the predictive information advantage of the fully recovered state over an imperfect return-based measure and motivates using 3PRF to extract the persistent state from Leader valuations.

4 Data and Recursive Empirical Design

4.1 Data

We use book-to-market ratios for the size- and value-sorted portfolios of Fama and French (1993), which divide U.S. stocks into 6, 25, or 100 portfolios, as in Kelly and Pruitt (2013). Files available on the website of Ken French report monthly portfolio market equity and annual portfolio book equity. We use these series to construct a monthly panel of book-to-market ratios. The ratio for month t uses total portfolio market capitalization at the end of month t and the latest observable annual total book equity. Following Fama and French (1993), we assume that book equity for calendar year Y becomes observable after June of year $Y + 1$. Daily returns for the same portfolios, also from the website of Ken French, are used to identify Leaders through LASSO.

The U.S. sample runs from January 1930 through December 2025, extending the sample of Kelly and Pruitt (2013) beyond its December 2010 endpoint. Market returns are measured by the CRSP value-weighted index, and the risk-free rate is the 30-day Treasury bill return.

4.2 Recursive Forecasts and Evaluation

We estimate the LASSO peer-return regressions over rolling five-year windows of daily returns and the subsequent 3PRF over expanding windows of monthly valuations and market returns. Following Kelly and Pruitt (2013), the formal out-of-sample evaluation begins with the market return for January 1980. Evaluation periods are dated by the realized return month: the forecast for January 1980 is formed at the end of December 1979.

For forecast origins $t = t_0, \ldots, T$, we measure predictive accuracy by

$$R^2_{OOS} = 1 - \frac{\sum_{t=t_0}^{T}(r_{m,t+1} - \widehat{r}_{m,t+1|t})^2}{\sum_{t=t_0}^{T}(r_{m,t+1} - \overline{r}_{m,t})^2}, \tag{23}$$

where $\overline{r}_{m,t}$ is the historical-mean benchmark, computed from all market returns observed through month t. A positive R^2_{OOS} indicates a reduction in mean squared forecast error relative to this benchmark. We report ENC-NEW significance bounds (Clark and McCracken 2001) and p-values for the adjusted-MSPE test of Clark and West (2007), using Newey–West

standard errors with 12 monthly lags and a moving-block bootstrap. Adapted from Sarno et al. (2012, Appendix F), the bootstrap imposes the null and jointly resamples forecast-level observations in twelve-month blocks to preserve their within-block dependence; Internet Appendix B provides the procedure. These tests assess the stated forecast comparisons, conditional on the constructed forecast histories. Their p-values are not adjusted for search across specifications, a distinction emphasized by the multiple-testing literature (Harvey et al. 2016).

We evaluate economic value for a mean–variance investor who allocates between the market and the risk-free asset (Campbell and Thompson 2008; Ferreira and Santa-Clara 2011). At the end of month t, the optimal equity weight, w_t, is

$$w_t = \frac{\widehat{r}_{t+1} - r_{f,t+1}}{\gamma \widehat{\sigma}^2_{t+1}}, \tag{24}$$

where $\widehat{r}_{t+1}$ is the out-of-sample market-return forecast, $r_{f,t+1}$ is the risk-free rate, and $\widehat{\sigma}^2_{t+1}$ is the variance of market returns estimated from all observations available through t. Following Neely et al. (2014), we restrict the equity weight to the interval from 0 to 1.5 ($0 \leq w_t \leq 1.5$), ruling out short selling and limiting leverage. The realized portfolio return is

$$r_{p,t+1} = w_t r_{m,t+1} + (1 - w_t) r_{f,t+1}.$$

The certainty-equivalent return and Sharpe ratio are

$$CER_p = \widehat{\mu}_p - \frac{\gamma}{2}\widehat{\sigma}^2_p, \qquad SR_p = \frac{\widehat{\mu}_p - \overline{r}_f}{\bar{\sigma}_p}, \tag{25}$$

where $\widehat{\mu}_p$ and $\widehat{\sigma}^2_p$ are the sample mean and variance of portfolio returns over the evaluation period, $\overline{r}_f$ is the corresponding mean risk-free rate, and $\bar{\sigma}_p$ is the standard deviation of excess portfolio returns. We set $\gamma = 2$ and report ΔCER and ΔSR, the annualized differences between the predictive strategy and the historical-mean strategy.

5 Main Results

Our main analysis focuses on the 100-portfolio panel. Its finer cross-section preserves greater heterogeneity in information precision and leaves more valuations for 3PRF estimation after Leader selection. The 6- and 25-portfolio panels contain substantially fewer cross-

sectional units, which may reduce the precision of factor estimates. Results for these coarser panels are reported as supplementary evidence in Internet Appendix C.3.

Table 1 reports 3PRF forecasts based on the full panel of present values and on valuations of Positive and Negative Leaders over two evaluation periods. Panel A covers January 1980–December 2010, the original out-of-sample evaluation period of Kelly and Pruitt (2013); Panel B extends the endpoint through December 2025. For comparison, we estimate predictive regressions using a Leader return series constructed from the Leader set selected in each month and the corresponding full-month returns, weighted by monthly market equity.

Table 1: **Main Out-of-Sample Forecasting Results**

Model	R^2_{OOS} (%)	p_{ENC}	p_{CW}	p_{MBB}	ΔCER	ΔSR
Panel A: January 1980–December 2010						
3PRF forecasts from present values						
Full Panel	0.925	< 0.05	0.012	0.014	2.731	0.197
Positive Leaders	−0.502	> 0.10	0.506	0.493	−0.594	−0.044
Negative Leaders	**2.614**	**< 0.01**	**0.000**	**0.001**	**4.179**	**0.323**
Predictive-regression forecasts from Leader returns						
Positive Leaders	1.322	< 0.01	0.048	0.025	2.574	0.241
Negative Leaders	0.585	< 0.05	0.125	0.103	1.123	0.100
Panel B: January 1980–December 2025						
3PRF forecasts from present values						
Full Panel	−0.191	< 0.10	0.271	0.271	0.001	0.006
Positive Leaders	−0.601	> 0.10	0.548	0.495	−0.532	−0.034
Negative Leaders	**1.540**	**< 0.01**	**0.003**	**0.001**	**2.188**	**0.168**
Predictive-regression forecasts from Leader returns						
Positive Leaders	−0.059	< 0.05	0.275	0.246	0.739	0.088
Negative Leaders	−0.089	> 0.10	0.345	0.316	0.186	0.034

Note: Valuation- and return-based forecasts use the same Leader sets. All tests are one-sided. p_{ENC} reports ENC-NEW significance bounds; p_{CW} and p_{MBB} report Clark–West p-values based on Newey–West standard errors with 12 monthly lags and a moving-block bootstrap, respectively. ΔCER and ΔSR are computed with $\gamma = 2$; ΔCER is reported in percentage points.

The present-value forecast based on Negative Leaders delivers an R^2_{OOS} of 2.614% over 1980–2010, compared with 0.925% for the full-panel estimator. Over 1980–2025, it retains an R^2_{OOS} of 1.540%, while the full-panel forecast falls to −0.191%. The forecast based on Negative Leaders is significant at the 1% level under all three tests in both periods. Annualized CER gains are 4.179 and 2.188 percentage points, with corresponding Sharpe ratio gains of

0.323 and 0.168. These results are consistent with Proposition 3.1: the negative tail more sharply separates assets by information precision, providing a more informative valuation panel for factor extraction.

The weaker performance of present-value forecasts based on Positive Leaders may reflect a trade-off between information quality and cross-sectional dimension. Selecting the top 10% of portfolios by return deviation leaves fewer valuations over which to average asset-specific noise. The improvement in information quality from positive-tail selection may not offset the cost of estimating the factor from a much smaller panel.

The return-based specifications show that Positive Leaders can nevertheless contain useful predictive information. Using the same Leader sets at each date, these specifications provide a portfolio-level analogue to Chen, Hao, Yu and Zhou (2024).[2] Over 1980–2010, the Positive Leader return predictor achieves an R^2_{OOS} of 1.322%, is significant under both Clark–West procedures, and generates annualized CER and Sharpe ratio gains of 2.574 percentage points and 0.241. In the extended sample, its R^2_{OOS} is slightly negative and the Clark–West tests no longer reject, although economic gains remain positive. Its stronger performance than the Negative Leader return predictor need not contradict Proposition 3.1, which ranks expected high-precision shares rather than the predictive content of Leader returns. The latter also depends on state exposure, current and delayed news responses, and asset-specific disturbances.

To examine how forecast performance evolves, define the cumulative sum of squared prediction error differences (CSSED) through forecast origin t as

$$\text{CSSED}_t = \sum_{\tau=t_0}^{t} \left[(e^{\text{HA}}_{\tau+1})^2 - (e^{\text{model}}_{\tau+1})^2 \right], \qquad t = t_0, \ldots, T.$$

An upward-sloping path indicates a reduction in squared prediction error relative to the historical mean.

Figure 1 shows that the full-panel 3PRF accumulates forecast gains during the original evaluation period but loses its cumulative advantage by the end of 2025. The decline is espe-

[2] Our implementation differs slightly in the network estimation window, the return horizon used to identify Leaders, and the construction of peer weights. These choices reflect our present-value model and empirical design. The return-based and 3PRF forecasts use identical Leader sets to ensure a fair comparison.

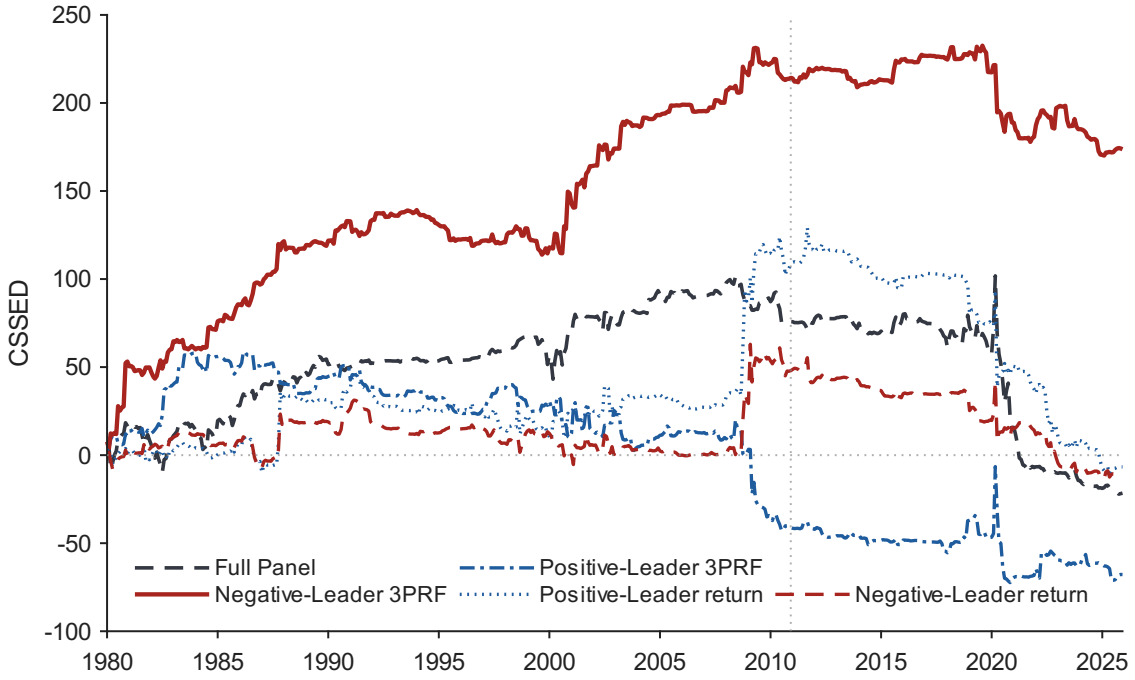


Figure 1: **Cumulative Out-of-Sample Forecast Performance.** Cumulative sums of squared prediction error differences are measured relative to the recursive historical mean. Positive values indicate lower cumulative squared prediction errors than the benchmark. The vertical dotted line marks December 2010.

Alt text: The Negative-Leader valuation forecast accumulates the largest gains and ends well above zero. Full-Panel gains fall below zero late in the sample; both Leader-return forecasts also finish slightly negative.

cially pronounced after 2020, indicating that December 2010 is the endpoint of the original sample rather than a discrete break in performance. In contrast, the CSSED of the 3PRF forecast based on Negative Leaders remains positive throughout the evaluation period and rises over multiple subperiods. This pattern indicates sustained cumulative outperformance relative to the historical mean, with recurring incremental gains and a substantial cumulative advantage at the end of 2025. Forecasts based on Leader returns also accumulate meaningful gains during the original period, especially for Positive Leaders. Their CSSED paths subsequently decline and finish slightly below zero in 2025, consistent with the negative extended-sample R^2_{OOS} values in Table 1.

6 Economic Mechanisms

The forecasting gains raise three questions about the information captured by Leader selection and its role in the cross-sectional present-value framework:

(i) *Do Negative Leaders respond more strongly to aggregate news than other portfolios?* If selection identifies more precise valuation measurements, selected assets should respond more strongly to independently measured news.

(ii) *Does dynamic Leader identification improve forecasting?* If the assets revealing greater information precision change over time, updating the Leader set should contribute to forecast gains.

(iii) *Why does the full-panel 3PRF forecast deteriorate?* When Negative Leader selection is concentrated in fewer portfolios, incomplete incorporation of aggregate news may introduce greater distortions into full-panel factor extraction.

We address these questions through a series of empirical tests. Internet Appendix D provides complementary evidence on portfolio characteristics and the economic interpretation of estimated expected returns. Together, these analyses help explain the mechanisms underlying the forecasting results.

6.1 Leaders and Aggregate News

We first provide an external validation of Leader identification. Under the persistent precision process in the model, portfolios selected on day $d-1$ retain an information advantage in expectation on day d (see Internet Appendix A.3). We therefore use external daily news proxies to test whether Negative Leaders respond more strongly than Positive Leaders and Nonleaders to aggregate news arriving on day d.

Holding the latest estimated month-end peer network fixed, we compute return deviations on each trading day $d-1$ and form the Negative and Positive Leader groups. Portfolios outside both tails form the Nonleader group. For $g \in \{-, +, N\}$, corresponding to Negative Leaders,

Positive Leaders, and Nonleaders, respectively, let $\mathcal{P}_{g,d-1}$ denote the group formed on day $d-1$. Its raw return and network-adjusted return deviation on the next trading day d are

$$r_{g,d} = \frac{1}{|\mathcal{P}_{g,d-1}|} \sum_{i \in \mathcal{P}_{g,d-1}} r_{i,d}, \qquad r_{g,d}^{\text{adj}} = \frac{1}{|\mathcal{P}_{g,d-1}|} \sum_{i \in \mathcal{P}_{g,d-1}} \left(r_{i,d} - \sum_{j \neq i} \widehat{w}_{i,j,d-1} r_{j,d} \right).$$

Here, $\widehat{w}_{i,j,d-1}$ is the latest month-end estimate of the peer-network weight available on day $d-1$. The network is therefore predetermined relative to returns on day d.

We consider five daily indicators of financial conditions. VIX measures option-implied S&P 500 volatility, while VVIX measures uncertainty about volatility itself. The U.S. contribution to the Office of Financial Research (OFR) Financial Stress Index measures domestic financial stress. The Baa–10Y spread, the difference between the Baa corporate bond yield reported by Moody's and the 10-year Treasury yield, captures credit-market conditions. The newspaper-based Overall Equity Market Volatility (EMV) tracker measures news coverage of equity-market volatility and uncertainty.[3] For each indicator X, we use the daily change as a proxy for aggregate news, treating decreases as favorable and increases as unfavorable:

$$\Delta X_d = X_d - X_{d-1}, \qquad G_d = \max(-\Delta X_d, 0), \qquad B_d = \max(\Delta X_d, 0).$$

For each comparison group $g \in \{N, +\}$, we use either $Y_d = r_{-,d} - r_{g,d}$ or $Y_d = r_{-,d}^{\text{adj}} - r_{g,d}^{\text{adj}}$, yielding four specifications. Raw returns capture differences in overall news sensitivity. Network-adjusted deviations measure news responses relative to the estimated peer portfolios and correspond more closely to the asset–peer comparison in Proposition 3.1. For each proxy and specification, we estimate

$$Y_d = \alpha + \beta_G G_d - \beta_B B_d + \varepsilon_d. \tag{26}$$

Positive coefficients $\beta_G > 0$ and $\beta_B > 0$ indicate that Negative Leaders respond more positively to favorable news and more negatively to unfavorable news, respectively. We test these one-sided alternatives using Newey–West standard errors with 21 trading-day lags.

Table 2 reports positive estimates of both β_G and β_B in every specification. Negative Leaders therefore tend to respond more positively to favorable news and more negatively to unfavorable news than either comparison group. For VIX, VVIX, and OFR financial stress,

[3] We obtain VIX and VVIX from Cboe, the U.S. financial stress contribution from the OFR, and Baa–10Y from FRED. Overall EMV is from the replication data of Baker, Bloom, Davis, and Kost.

Table 2: **Return Responses to Aggregate News across Portfolio Groups**

	Raw returns		Network-adjusted return deviations	
	(1) Negative–Nonleader	(2) Negative–Positive	(3) Negative–Nonleader	(4) Negative–Positive
Panel A: VIX (January 3, 1990–December 31, 2025)				
Good news (β_G)	12.336***	11.657***	21.033***	21.899***
	(1.933)	(2.771)	(2.369)	(2.975)
Bad news (β_B)	11.448***	8.044***	15.587***	8.625***
	(1.607)	(1.731)	(2.250)	(2.684)
Panel B: VVIX (March 16, 2006–December 31, 2025)				
Good news (β_G)	4.810***	4.857**	12.360***	13.483***
	(1.957)	(2.215)	(2.137)	(2.805)
Bad news (β_B)	6.726***	5.822***	8.841***	5.550*
	(1.338)	(1.976)	(1.923)	(3.510)
Panel C: OFR U.S. Financial Stress (January 4, 2000–December 31, 2025)				
Good news (β_G)	13.868***	6.981**	21.750***	14.475***
	(2.157)	(3.109)	(3.083)	(4.536)
Bad news (β_B)	14.842***	11.341***	19.490***	12.665***
	(1.941)	(2.455)	(3.019)	(3.754)
Panel D: Baa–10Y Spread (January 3, 1986–December 31, 2025)				
Good news (β_G)	2.227**	1.250	1.591*	0.142
	(1.328)	(1.425)	(1.069)	(1.327)
Bad news (β_B)	2.813	2.748	5.798**	5.275*
	(3.869)	(3.933)	(2.891)	(3.256)
Panel E: Overall EMV (January 3, 1985–March 14, 2024)				
Good news (β_G)	2.094**	0.979	2.649***	2.764**
	(1.085)	(1.123)	(0.950)	(1.346)
Bad news (β_B)	3.114**	2.690*	4.022***	4.239**
	(1.618)	(1.673)	(1.643)	(1.993)

Note: Dependent variables are outcomes for Negative Leaders minus those for the comparison group. Sample periods are reported in the panel headings and depend on data availability. Coefficients are reported in basis points per standard deviation of ΔX. Parentheses contain Newey–West standard errors with 21 trading-day lags. *, **, and *** denote significance at the 10%, 5%, and 1% levels in one-sided tests of $\beta_G > 0$ and $\beta_B > 0$.

most coefficients are significant at the 1% level. Across all five proxies, the few statistically insignificant estimates are also positive. For example, a one-standard-deviation decline in VIX is associated with an increase of 12.336 basis points in the raw return spread between Negative Leaders and Nonleaders; an increase of the same size is associated with a decline of 11.448 basis points.

These results support the selection mechanism in Proposition 3.1. Proxies constructed independently of selection and forecasting provide external evidence on the news sensitivities of selected portfolios. This evidence bears more directly on selection than out-of-sample forecast performance alone, which also reflects factor estimation and the subsequent predictive regression. Differences in economic content and sample coverage can account for variation in coefficient magnitude and precision across proxies. Despite this variation, the Negative–

Nonleader coefficients generally exceed the corresponding Negative–Positive coefficients. The point estimates suggest that aggregate news sensitivity is greatest among Negative Leaders, followed by Positive Leaders and Nonleaders. Both raw returns and network-adjusted deviations are therefore consistent with the two tails identifying news-sensitive portfolios and the negative tail providing sharper separation.

6.2 Time-Varying Leader Selection

Information precision and state exposure affect selection probabilities, but realized Leader membership also depends on current news. The two Leader groups have similar market betas and return volatilities, while Negative Leaders differ significantly from Nonleaders in most reported characteristics; details appear in Internet Appendix D.1. These average characteristics coexist with substantial membership changes: average monthly turnover is 87.8% for Negative Leaders and 86.9% for Positive Leaders.

Figure 2 complements the turnover measures with selection frequencies over January 1980–December 2025 for the 100 size–book-to-market portfolios, displayed on the underlying 10×10 sort. Selection is broadly distributed across the cross-section, although frequencies tend to be higher in the highest book-to-market decile. The heat maps describe average selection probabilities, not permanent Leader types, consistent with the distinction in the model between shifting information advantages and realized Leader membership.

To assess whether fixed size or value subsets can match the forecast gains from dynamic selection, we compare dynamic identification of Negative Leaders with four fixed-subset strategies: the largest and smallest size deciles and the highest and lowest book-to-market deciles. Each subset contains ten portfolios, and all forecasts use the same recursive 3PRF procedure.

Table 3 shows that size-based strategies deliver at most modest predictive gains, while both book-to-market strategies have negative R^2_{OOS}. All four fixed subsets underperform dynamic identification of Negative Leaders in forecast accuracy, CER gains, and Sharpe ratio gains. The results support the importance of dynamic selection beyond fixed size or value membership.

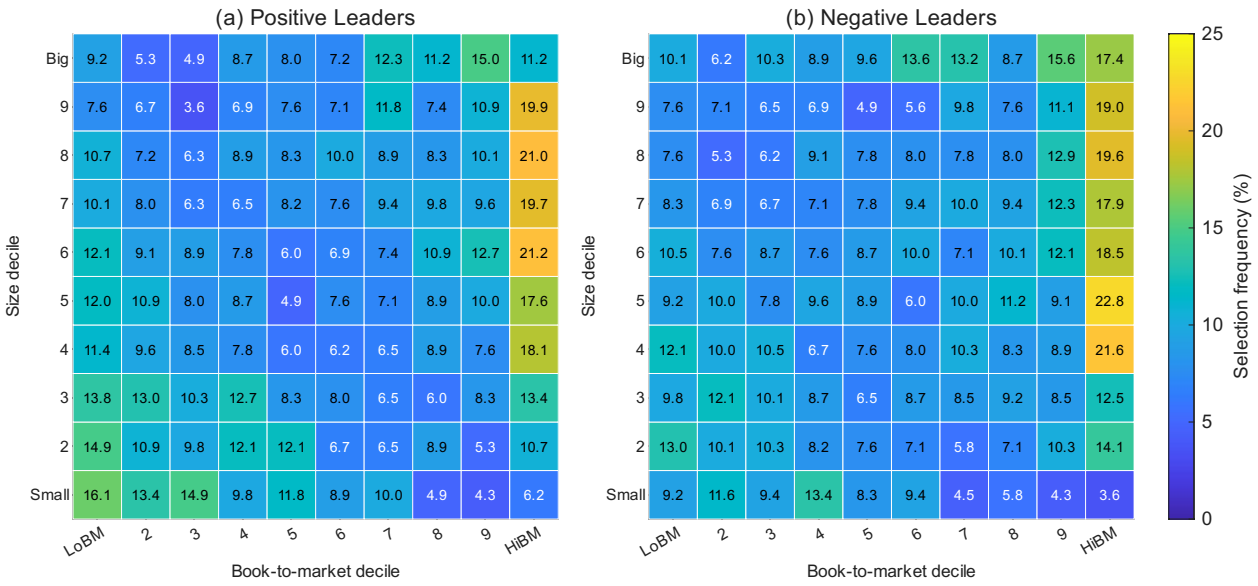


Figure 2: **Leader Selection Frequencies.** Each cell reports the percentage of months from January 1980 through December 2025 in which a size–book-to-market portfolio belongs to the indicated Leader set. Size increases from bottom to top, and book-to-market increases from left to right.

Alt text: Panel a: Positive-Leader selection frequencies are highest mostly in the rightmost book-to-market column. Panel b: Negative-Leader frequencies also cluster in the highest book-to-market column, particularly among middle-sized portfolios.

Table 3: **Fixed-Subset Strategies versus Dynamic Leader Identification**

Strategy	R^2_{OOS} (%)	p_{ENC}	p_{CW}	p_{MBB}	ΔCER	ΔSR
Largest-size decile	0.375	< 0.05	0.063	0.056	1.479	0.059
Smallest-size decile	−0.008	> 0.10	0.367	0.313	0.158	0.005
Highest-book-to-market decile	−1.775	> 0.10	0.669	0.681	−0.743	−0.017
Lowest-book-to-market decile	−0.244	> 0.10	0.859	0.832	−0.366	−0.030
Negative-Leader identification	**1.540**	**< 0.01**	**0.003**	**0.001**	**2.188**	**0.168**

Note: The sample covers January 1980–December 2025. Each fixed subset contains ten size–book-to-market portfolios. Forecast tests and economic gains follow Table 1.

We assess the information in the return-based selection rule with a placebo test using 1,000 forecast paths based on random monthly selection. At each monthly forecast origin, each path independently draws ten portfolios without replacement from the full universe of 100 and applies the same recursive 3PRF procedure. Figure 3 shows the resulting distribution of R^2_{OOS} over January 1980–December 2025. The median placebo R^2_{OOS} is −0.545%, whereas the forecast based on Negative Leaders achieves 1.540% and outperforms 999 of the 1,000 placebo forecasts. The advantage over randomly updated subsets of the same size supports

the predictive value of the return-based selection rule.

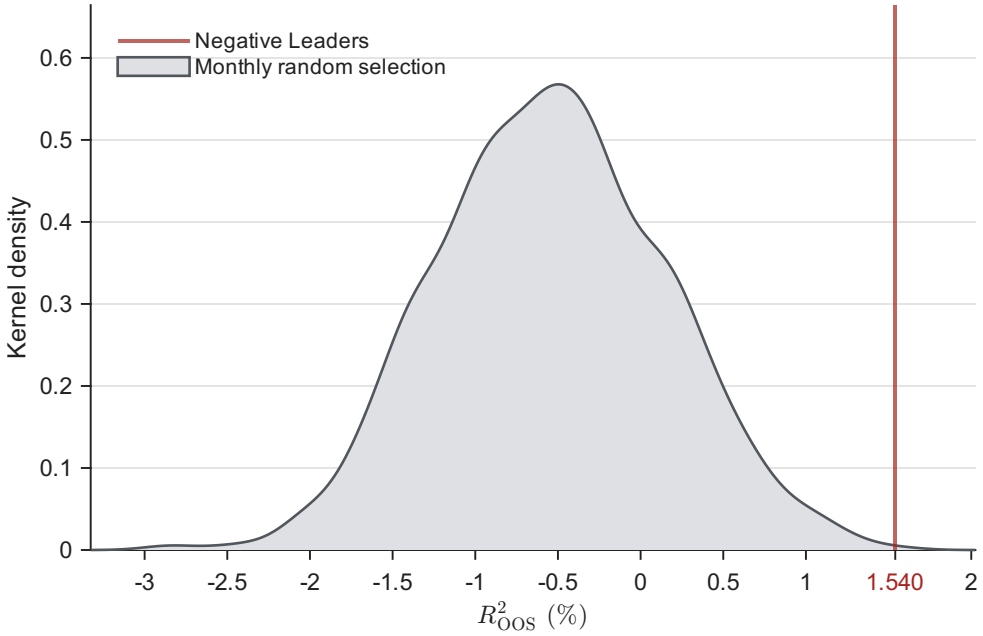


Figure 3: **Random-Subset Placebo Test.** The distribution of out-of-sample R^2 over January 1980–December 2025 is based on 1,000 random-selection forecast paths. Each path draws ten portfolios without replacement from the full universe of 100 at every monthly forecast origin. The vertical line marks the forecast based on Negative Leaders.

Alt text: The random-selection density peaks below zero. The Negative-Leader forecast lies in its far right tail, indicating stronger predictive accuracy than almost all random-selection paths.

6.3 Full-Panel 3PRF Forecast Deterioration

Dynamic Leader selection improves forecasting, but broadly distributed selection frequencies over the full sample can mask episodes in which evidence of timely information incorporation is concentrated in a narrower group of portfolios. Such concentration may leave full-panel factor extraction more exposed to common valuation errors: adding valuations that incorporate news less precisely need not produce proportionate information gains. We investigate this information concentration channel by relating full-panel forecast losses to the effective breadth of Negative Leader selection.

Let $n_{i,t}$ count the selections of portfolio i as a Negative Leader over months $t-59$ through t. Its share of all selections in the window is

$$p_{i,t}^{(60)} = \frac{n_{i,t}}{\sum_j n_{j,t}}.$$

The denominator accommodates changes in the number of eligible portfolios across months. A larger share indicates more frequent selection relative to other portfolios. We define the effective Leader pool as

$$N_t^{Eff} = \left[\sum_i \left(p_{i,t}^{(60)} \right)^2 \right]^{-1}. \tag{27}$$

This inverse-Herfindahl measure gives the number of portfolios that would generate the same concentration if selected equally often. Selecting the same ten portfolios each month gives $N_t^{Eff} = 10$; equal selection across all 100 gives $N_t^{Eff} = 100$. Unequal frequencies make the effective pool smaller than the number of distinct selected portfolios, so lower values indicate greater concentration. Guided by the identification result and external news evidence, we interpret the measure as an indicator of the breadth of revealed information advantages across portfolios, reflected in selection frequencies over the rolling window.

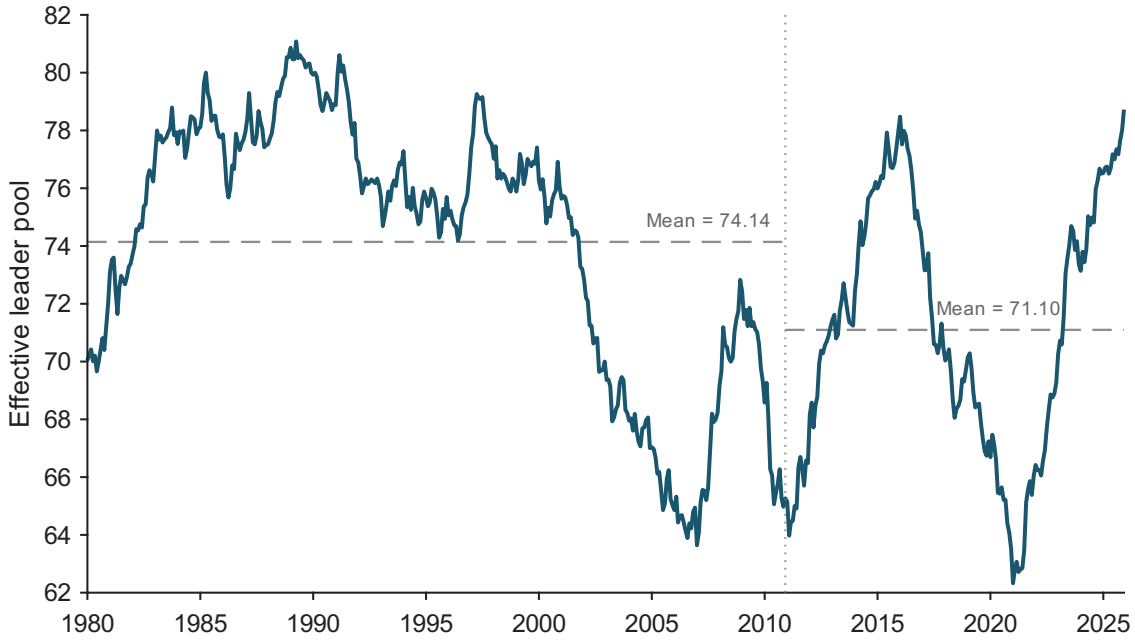


Figure 4: **Effective Pool of Negative Leaders.** The effective pool is computed from rolling 60-month selection shares. The vertical line marks December 2010. Horizontal dashed lines show the means for the two subperiods.

Alt text: Repeated declines are followed by recoveries; the deepest trough occurs late in the sample. The later subperiod has a lower average effective Leader pool, despite a strong final recovery.

Figure 4 shows that the average pool declines from 74.14 over 1980–2010 to 71.10 over 2011–2025, with pronounced contractions before the global financial crisis and around 2020.

The smaller pool in the later sample indicates more concentrated selection of Negative Leaders. To assess the information concentration interpretation, we test whether a smaller effective pool is associated with larger full-panel forecast losses relative to the historical-mean and Negative-Leader benchmarks:

$$(e_{t+1}^{All})^2 - (e_{t+1}^{j})^2 = a + \beta N_t^{Eff} + \boldsymbol{\delta}'\mathbf{Z}_t + \varepsilon_{t+1}, \tag{28}$$

where $j \in \{HA, NL\}$ denotes the historical-mean or Negative-Leader 3PRF forecast, respectively, and e_{t+1}^{j} is the market-return forecast error for method j. The vector $\mathbf{Z}_t$ contains the controls in each specification. Market volatility is the annualized standard deviation of monthly market log returns from $t-11$ through t. The Baa–10Y credit spread is the difference between the Baa corporate bond yield reported by Moody's and the 10-year Treasury yield in month $t-1$. The Fama–French three-factor controls (FF3) are the market excess return, SMB, and HML in month t.

Table 4 reports negative coefficients on the effective pool for both benchmarks in every specification, each significant at the 10% level in a one-sided test. Greater concentration of Negative Leader selection is therefore associated with larger full-panel forecast losses relative to both benchmarks, including after controlling for market volatility, credit conditions, and standard return factors. These findings support the information concentration channel: full-panel forecasts perform worse when revealed information advantages are less broadly distributed, consistent with common updating errors becoming relatively more important for factor extraction. Together with the smaller average effective pool after 2010, this evidence helps explain the deterioration of the full-panel 3PRF forecast in the extended sample.

7 Robustness

This section examines robustness across evaluation periods and economic states, trading costs and portfolio-allocation choices, learning algorithms, and model specifications, followed by supplementary evidence from coarser panels.

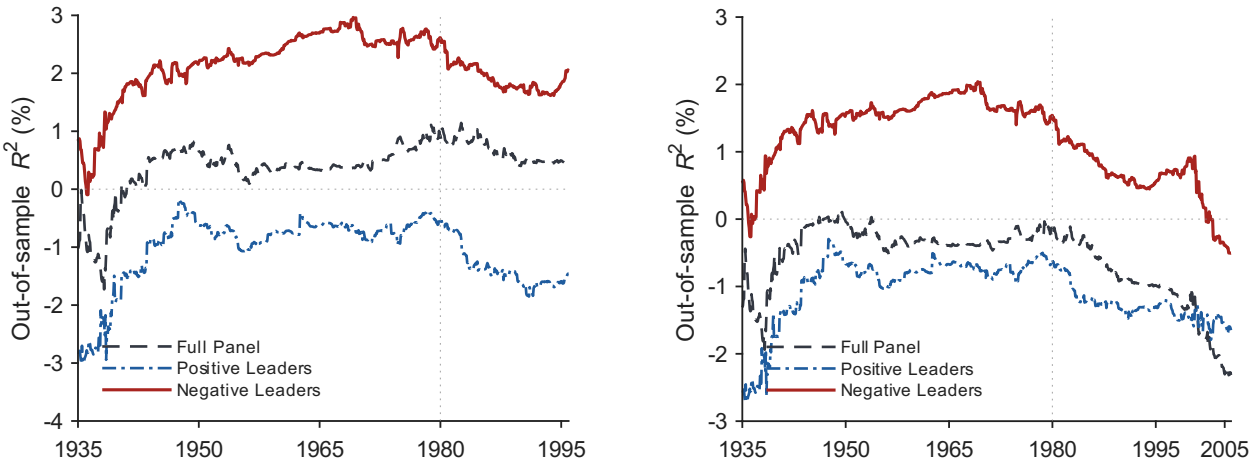
Table 4: **Effective Leader Pool and Full-Panel Forecast Losses**

Panel A: $(e_{t+1}^{All})^2 - (e_{t+1}^{HA})^2$	(1)	(2)	(3)	(4)
Effective Leader pool	-0.371^{**}	-0.247^{*}	-0.360^{**}	-0.233^{*}
	(0.208)	(0.151)	(0.210)	(0.152)
Market volatility		0.212		0.202
		(0.269)		(0.273)
Credit spread		0.229^{*}		0.237^{*}
		(0.135)		(0.130)
Market excess return			-0.221^{**}	-0.230^{**}
			(0.108)	(0.110)
SMB			0.142	0.132
			(0.187)	(0.192)
HML			-0.124	-0.111
			(0.124)	(0.125)
Constant	0.039	0.039	0.039	0.039
	(0.142)	(0.131)	(0.146)	(0.135)
Observations	552	552	552	552
R^2	0.011	0.018	0.017	0.023
Adjusted R^2	0.009	0.012	0.009	0.012
Panel B: $(e_{t+1}^{All})^2 - (e_{t+1}^{NL})^2$	(1)	(2)	(3)	(4)
Effective Leader pool	-0.206^{*}	-0.299^{**}	-0.211^{*}	-0.293^{**}
	(0.136)	(0.169)	(0.139)	(0.172)
Market volatility		0.144		0.144
		(0.178)		(0.181)
Credit spread		-0.176		-0.155
		(0.135)		(0.141)
Market excess return			-0.056	-0.048
			(0.235)	(0.236)
SMB			0.058	0.038
			(0.154)	(0.159)
HML			0.202	0.192
			(0.189)	(0.192)
Constant	0.354^{***}	0.354^{***}	0.354^{***}	0.354^{***}
	(0.113)	(0.088)	(0.109)	(0.088)
Observations	552	552	552	552
R^2	0.003	0.006	0.006	0.009
Adjusted R^2	0.001	0.001	-0.001	-0.002

Note: The sample covers January 1980–December 2025. The dependent variables are differences in squared monthly forecast errors, measured in squared percentage points. Explanatory variables are standardized. Parentheses contain Newey–West standard errors with 60 monthly lags. *, **, and *** denote significance at the 10%, 5%, and 1% levels. Tests are one-sided for $\beta < 0$ on the effective pool and two-sided for the other coefficients.

7.1 Evaluation Periods and Economic States

Figure 5 varies the first evaluation month while retaining the same recursive forecast history, with endpoints in December 2010 and December 2025. Earlier starting dates include more forecasts based on shorter estimation histories, whereas later dates focus on a shorter, more recent evaluation period. Negative-Leader forecasts outperform the Full Panel and Positive Leader forecasts across a broad range of starting dates in both samples, indicating that their relative advantage does not depend on the January 1980 starting date. In the extended sample, absolute performance weakens as evaluation begins later, but the advantage over the other two forecasts remains. Internet Appendix C.1 provides complementary evidence from rolling twenty-year evaluation windows.



(a) 100 portfolios, ending 2010

(b) 100 portfolios, ending 2025

Figure 5: **Out-of-Sample R^2 across Forecast Starting Dates.** Out-of-sample R^2 is plotted against the first evaluation month. The vertical line marks January 1980.

Alt text: Panel a: Negative-Leader forecasts remain above the other curves across starting dates, with mostly positive forecasting gains. Panel b: Negative Leaders retain their relative advantage, but all forecasts fall below zero for the latest starting dates.

Table 5 evaluates forecasts after separately excluding the global financial crisis and the COVID-19 episode, and across economic states. Forecasts based on Negative Leaders deliver positive R^2_{OOS}, CER gains, and Sharpe ratio gains that exceed the corresponding Full Panel estimates in every subsample. Predictability remains significant after either crisis is excluded and in most economic states. The forecasting advantage is therefore not confined to crises or

a particular economic state.

Table 5: **Forecast Performance across Subsamples**

	Negative Leaders			Full Panel		
Evaluation sample	R^2_{OOS}	ΔCER	ΔSR	R^2_{OOS}	ΔCER	ΔSR
Panel A: Excluding crisis periods						
Exclude GFC	1.404***	1.620	0.126	−0.129	0.150	0.032
Exclude COVID-19	1.913***	2.638	0.203	0.200*	0.402	0.025
Panel B: Economic states						
Recession	1.385	4.551	0.177	−2.051	−3.076	−0.125
Expansion	1.584***	1.911	0.146	0.329*	0.366	0.047
High volatility	0.827*	1.785	0.105	−0.962	−1.788	−0.104
Low volatility	3.577***	2.591	0.318	2.008**	1.797	0.245
Tightening	0.981**	0.855	0.079	−0.037	0.382	0.047
Easing	1.997***	3.450	0.235	−0.317	−0.358	−0.020

Note: The sample covers January 1980–December 2025. Panel A excludes July 2007–June 2009 (GFC) or February–December 2020 (COVID-19). Panel B classifies states at the forecast origin using NBER recessions, market variance above its sample median, and monetary tightening (easing), defined by a federal funds rate above (at or below) its level twelve months earlier. R^2_{OOS} is reported in percent; economic-value settings and units follow Table 1. Stars on R^2_{OOS} indicate significance in the one-sided Clark–West test with Newey–West standard errors using 12 monthly lags: $^*p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$. Economic gains summarize returns within each subsample using the original allocations.

7.2 Economic Value and Trading Costs

Statistical accuracy and economic value need not rank forecasts identically. Squared forecast errors determine R^2_{OOS}, whereas economic gains depend on the allocations induced by forecast signs and magnitudes, the variance estimate, and risk aversion. Weight constraints further limit the portfolio response to extreme forecasts. Holding the forecasts fixed, Table 6 examines how risk aversion, trading costs, and portfolio weight constraints affect economic gains for Negative Leaders over January 1980–December 2025.

Forecasts based on Negative Leaders retain positive CER and Sharpe ratio gains in every specification. The gains decline as trading costs rise but remain positive at 30 basis points, including with unrestricted equity weights. Higher risk aversion reduces turnover. By contrast, the Full Panel and Positive Leader forecasts yield negative economic gains once trading costs are included; these results are omitted for brevity.

Table 6: **Economic Value and Trading Costs for Negative Leaders**

γ	Turnover (%)	0 bps		15 bps		30 bps	
		ΔCER	ΔSR	ΔCER	ΔSR	ΔCER	ΔSR
Panel A: Baseline allocation bounds ($0 \leq w \leq 1.5$)							
2	494.8	2.188	0.168	1.524	0.118	0.860	0.068
3	359.7	1.669	0.174	1.180	0.122	0.691	0.071
5	220.5	1.040	0.178	0.743	0.127	0.445	0.075
Panel B: No leverage cap ($w \geq 0$)							
2	548.9	2.617	0.178	1.878	0.127	1.138	0.076
3	366.7	1.741	0.178	1.243	0.127	0.745	0.075
5	220.5	1.040	0.178	0.743	0.127	0.445	0.075
Panel C: Unrestricted weights ($w \in \mathbb{R}$)							
2	676.2	3.051	0.193	2.150	0.134	1.248	0.075
3	451.5	2.028	0.193	1.422	0.134	0.815	0.074
5	271.4	1.209	0.193	0.847	0.134	0.484	0.074

Note: The sample covers January 1980–December 2025. ΔCER is reported in percentage points. Turnover is annualized and reported in percent. Both strategies incur the stated costs per unit of one-way turnover.

7.3 Alternative Machine-Learning Methods

We examine sensitivity to the learning methods used in both stages. Panel A of Table 7 replaces LASSO with OLS, ridge regression, or the elastic net, retaining the coefficient normalization defined after equation (17). Panel B uses nonlinear models: random forests, gradient-boosted trees, and neural networks. Following Gu et al. (2020), we consider three network architectures. NN1 has one hidden layer with 32 neurons; NN2 has two hidden layers with 32 and 16 neurons; NN3 has three hidden layers with 32, 16, and 8 neurons. These models allow interactions among contemporaneous peer returns. Their selection signal is

$$RetDev_{i,t}^{\mathrm{ML}} = r_{i,t}^{\mathrm{last}} - \widehat{f}_{i,t}\left(\mathbf{r}_{-i,t}^{\mathrm{last}}\right),$$

where $\widehat{f}_{i,t}$ predicts the return on asset i from returns on the other portfolios. Internet Appendix E provides the model specifications and tuning procedure.

All first-stage alternatives retain lower-decile selection and the baseline 3PRF forecasting procedure. Panel C instead retains the baseline Leader sets and replaces 3PRF with one or three principal components extracted from their standardized valuation histories. These components maximize explained valuation variance without targeting subsequent market

returns. Panel D reports the baseline for comparison.

Table 7: **Alternative Machine-Learning Methods**

Method	R^2_{OOS} (%)	p_{ENC}	p_{CW}	p_{MBB}	ΔCER	ΔSR
Panel A: Alternative linear peer-return models						
OLS	0.500	< 0.01	0.036	0.032	1.573	0.104
Ridge	1.261	< 0.01	0.004	0.002	2.443	0.182
Elastic-net	1.219	< 0.01	0.014	0.005	1.774	0.136
Panel B: Nonlinear peer-return models						
Random Forest	1.172	< 0.01	0.002	0.003	2.364	0.186
Gradient-Boosted Trees	1.185	< 0.01	0.005	0.006	2.640	0.190
NN1 (32)	1.515	< 0.01	0.019	0.015	2.544	0.204
NN2 (32-16)	1.410	< 0.01	0.007	0.006	2.907	0.236
NN3 (32-16-8)	0.924	< 0.01	0.022	0.033	2.019	0.173
Panel C: Alternative factor extraction						
PCA, one factor	-0.576	> 0.10	0.800	0.846	-1.954	-0.132
PCA, three factors	-0.683	> 0.10	0.323	0.337	-1.848	-0.117
Panel D: Baseline						
LASSO + 3PRF	1.540	< 0.01	0.003	0.001	2.188	0.168

Note: The sample covers January 1980–December 2025. All forecasts use Negative Leaders. All tests are one-sided. p_{ENC} reports ENC-NEW significance bounds; p_{CW} and p_{MBB} report Clark–West p-values based on Newey–West standard errors with 12 monthly lags and a moving-block bootstrap, respectively. ΔCER and ΔSR are computed with $\gamma = 2$; ΔCER is reported in percentage points.

All linear and nonlinear peer models retain positive R^2_{OOS} and economic gains, with predictability significant at the 5% level under all three tests. NN1 achieves forecast accuracy close to the baseline, and both tree models, NN1, and NN2 deliver larger CER and Sharpe-ratio gains. In contrast, both PCA specifications yield negative R^2_{OOS} and economic gains, highlighting the importance of supervised factor extraction. These findings show that the framework's forecasting benefits extend across learning algorithms: economically guided Leader selection can be combined with different peer-return models while retaining target-guided factor extraction.

7.4 Alternative Specifications

We examine alternative return-deviation signals and treatments of network coefficients; definitions and complete results appear in Internet Appendix C.2. Exponentially weighted

averages, the five-day simple average, and volatility scaling retain positive predictability and economic gains, whereas the ten-day simple average, cumulative monthly deviations, and monthly extremes yield negative R^2_{OOS}. None of these alternative signals improves predictive accuracy relative to the final-day baseline. Both post-estimation truncation and nonnegative LASSO retain significant predictability and positive economic gains but underperform the unrestricted baseline. These comparisons underscore the importance of recent deviations and suggest that negative conditional return links provide useful information for identifying Leaders.

Alternative network estimation windows retain positive predictability and economic gains. Additional factors extracted using the automatic-proxy procedure of Kelly and Pruitt (2015) reduce predictive accuracy: R^2_{OOS} remains positive with two factors but turns negative with three, while economic gains remain positive in both cases. Selecting too few portfolios (5%) may reduce factor estimation precision, whereas selecting too many (30%) may dilute information quality; both specifications yield negative R^2_{OOS}. These comparisons show that forecast gains persist across network estimation windows, while the factor and tail results illustrate the balance between information quality and factor estimation precision.

7.5 Portfolio Granularity

Finally, the 6- and 25-portfolio panels provide supplementary evidence on Leader selection with coarse aggregation and small cross-sections. Internet Appendix C.3 reports the detailed results. Aggregation may obscure differences in information precision, while Leader tails of 50% and 20% leave only about three and five valuations for factor estimation, respectively, increasing exposure to estimation error. Despite these constraints, Leader selection improves out-of-sample forecast accuracy relative to the Full Panel in the 25-portfolio panel over both evaluation periods.

8 Conclusion

This paper combines return-based Leader identification with supervised factor extraction from leading present values. Our model incorporates heterogeneous, time-varying information precision and ambiguity that induces stronger updating in response to unfavorable signals. Incomplete news incorporation can generate common valuation errors that violate the weak-dependence assumption underlying 3PRF consistency, motivating the selection of more precise measurements. We establish sufficient conditions under which negative-tail selection identifies a higher expected proportion of currently high-precision assets than random or positive-tail selection. Simulations with rolling LASSO estimation illustrate these gains in finite panels.

Forecasts based on Negative Leader valuations deliver significant out-of-sample gains and positive economic value, outperforming the Full Panel and comparable Leader return forecasts. Forecast gains persist across economic states and linear and nonlinear identification models, while economic value survives trading costs. External news measures support the selection mechanism, while comparisons with fixed subsets and random monthly selection support dynamic identification. Greater concentration of Negative Leader selection is associated with larger full-panel forecast losses, helping explain deterioration in the extended sample.

More broadly, the framework illustrates how economic structure can guide supervised representation learning in asset pricing. Machine learning estimates cross-sectional return relationships, the information model guides valuation selection, and the forecasting target guides factor extraction. This approach extends the role of machine learning beyond direct return prediction to the identification and extraction of economically informative signals.

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